

TECH TRACK · PROJECT BRIEF

Mini Project 1: pattern-based workflow

The Dhaga & Co. Engagement



You are being given a client, not a problem statement. Find the problem worth solving, build a working MVP that solves it, and present it back to the client.

FORMAT

Groups of five

DELIVERABLE

A deployed MVP with a frontend

PHASES

Discover, build, present

SUBMISSION

One per group, presented live

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How this project works

You work in groups of five. One submission per group. Three phases, in order.

1. Discover	2. Build	3. Present
Read the client. Decide which problem is worth solving and prove why. Write it down before you write any code.	Build an MVP that solves the problem you chose. It must have a frontend and it must be deployed to a URL.	Present the MVP to the client. The problem, why it matters, a live demo, and what you would build next.

There is no problem statement in this brief. What follows is a client, described the way a client actually arrives: a business with revenue, people, systems, habits and complaints, and no agreement about what to fix first. Deciding what to fix is the first half of the work.

Working as a five

Split the work however you want, but two things are not negotiable. Everyone must be able to explain the whole system, not only the part they touched. And every member presents at least one segment of the final pitch.

A WARNING ABOUT GROUP SHAPE

The most common failure in a five is one person building and four people watching. The second most common is five people building five disconnected things in parallel. Decide early who owns the discovery write-up, who owns the frontend, who owns the workflow, who owns deployment, and who owns the pitch, then trade work across those lines deliberately.

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The client: Dhaga & Co.

Dhaga & Co. is a direct-to-consumer fashion brand, founded in 2019 in Bengaluru by two former merchandisers who left a large retail chain to sell affordable everyday clothing online.

What they sell

Womenswear is sixty percent of revenue, kidswear thirty, men's basics the rest. Prices run from ₹399 to ₹1,499, and the average order value is ₹840. New collections drop every Tuesday and Friday, roughly four hundred new SKUs a week, with about fourteen thousand SKUs live at any moment. The catalogue turns over fast: a product that has not sold in six weeks is pulled.

Who buys from them

Customers are eighteen to thirty-four, seventy-eight percent women, and sixty-four percent live in tier-2 and tier-3 cities. Ninety-two percent of orders come through the Android app, often on low-end handsets and patchy connections. Sixty-one percent of orders are cash on delivery. Customers search in Hinglish, and they search by occasion rather than by product name: "mehndi function dress", "office wear kurti", "school ke liye shirt".

Scale

Around ₹310 crore GMV on a run-rate basis, about forty-eight thousand orders a week, 3.1 million app installs and roughly seven hundred thousand monthly active users. The company employs 190 people, sixteen of them engineers.

ABOUT THIS CLIENT

Dhaga & Co. is a composite, built from how businesses of this kind actually run. Treat every number in this brief as real and verified. Anything not stated here is something you would have to ask the client for, and saying so out loud is a better answer than quietly inventing it.

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How they operate today

Function	How it runs right now
Sourcing	Around forty vendor partners, mostly in Tiruppur and Jaipur. Purchase orders are raised in spreadsheets and confirmed over WhatsApp. Lead times vary by vendor and are held in people's heads.
Cataloguing	A sample arrives, goes through a studio shoot, then a six-person listing team types attributes into an internal admin and writes the product copy by hand. Sample to live takes six to nine days.
Warehousing	Three fulfilment centres, in Bhiwandi, Gurugram and Hyderabad, running on Unicommerce. Stock is allocated to the nearest centre with inventory.
Delivery	Split across Delhivery, Shiprocket and Ekart. Four to seven days is typical, longer into the north east.
Returns	Raised by the customer in the app or over WhatsApp, collected by reverse pickup, inspected at the fulfilment centre. Refunds go out after inspection.
Support	Thirty-four agents on Freshdesk, plus a WhatsApp line running through Gupshup. Agents work from a shared document of canned replies.

Function	How it runs right now
Merchandising	Two merchandisers set the homepage and the order of every category page by hand, each morning, based on yesterday's sales and instinct.
Marketing	Meta and Google performance ads, influencer seeding, and CleverTap for push and WhatsApp campaigns.
Analytics	Metabase sitting on a Postgres read replica, with two analysts serving the whole company.

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What their data actually looks like

Where it lives	What is in it	Condition
Orders	Eleven million rows in Postgres, with line items, payment mode, address and status history.	Clean and trustworthy.
Catalogue	Fourteen thousand live SKUs with sixty-odd attributes each, plus images.	Colour has been typed about ninety different ways. Fabric is free text. Size charts differ per vendor.
Returns	Reason captured as a dropdown with an "Other" free-text box.	Forty-four percent of returns land in "Other".
Support tickets	Roughly nine thousand a week in Freshdesk, mostly free text.	No consistent tagging. Agents pick a category only when they remember to.
Reviews	410,000 product reviews, star rating plus free text, eighteen months deep.	Nobody reads them. They are displayed and never analysed.
App events	Mixpanel, with search queries, taps, filters and scroll depth, retained eighteen months.	Complete, but only ever queried for funnels.
Vendor POs	One Google Sheet per vendor, plus WhatsApp threads.	Inconsistent columns. No single view across vendors.
Product copy	Around two hundred descriptions written by hand each week by four people.	Tone drifts between writers. Older listings were never revisited.

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What the people inside the company say

These are the things you would hear in your first week if you sat with each of them. They do not agree with each other about what matters most, and nobody has ranked this list.

Who	What they told you
Ritu, co-founder and CEO	"We acquire well and we retain badly. Repeat purchase rate has been stuck at twenty-two percent for six quarters. Every quarter we buy the same customer twice."
Sameer, Head of Growth	"Customer acquisition cost is up forty percent year on year. We cannot outspend our way out of this one."
Faizan, Head of Supply Chain	"Return to origin on cash on delivery is twenty-six percent. Each one costs us about ₹120 in logistics and burns a delivery slot we could have used."
Neha, Category Head	"Returns are thirty-one percent overall. When I read the Other box by hand, most of it is about fit, but I can only read a few hundred at a time."
Arpita, Head of CX	"Fifty-eight percent of tickets are some version of where is my order. My agents copy and paste the same four replies all day. Average first response is nine hours."
Vivek, Listing lead	"Six to nine days from sample to live. The drop calendar slips most weeks, and when it slips we lose the Tuesday traffic spike entirely."
Karthik, Data analyst	"Metabase tells us what happened. It never tells us why. Requests from the business queue up for about two weeks before I get to them."
Dev, CTO	"Sixteen engineers, none of them an ML engineer. Whatever you build, somebody here has to run it on the Monday after you leave."

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Constraints you have to build within

- Customers are mobile-first, on low-end Android handsets and unreliable connections.
- Hinglish and vernacular input is normal, not an edge case.
- Cash on delivery dominates, which changes the economics of everything downstream.
- There is no ML engineer. Anything you propose has to be operable by a small, busy team.
- Cost per action matters at forty-eight thousand orders a week. A rupee is a rupee.

- Anything customer-facing either has to be safe to publish unread, or has to have a human review step that you designed on purpose.

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Phase 1: Discovery

Nobody is going to tell you what to build. Deciding that, and being able to defend the decision, is the first thing you are being assessed on.

There is more than one defensible problem in the pages above. There are also plenty of indefensible ones. The difference is not how interesting the technology is. It is whether a named person inside Dhaga & Co. is losing something measurable today because the problem is unsolved.

The discovery note

One page, written and agreed by the group **before you write any code**. It has to contain:

- **The problem in one sentence**, in the client's language, not in yours.
- **Who owns it today** inside the company, and what they currently do instead.
- **The evidence** from the case study that took you there, cited specifically rather than gestured at.
- **What it costs them**, expressed in money, time, or a metric they already track and already argue about.
- **What success looks like**, and how you would measure it with data they already hold.
- **A ranked shortlist of at least four problems**, in order, with the reason each one sits where it does.
- **Your biggest assumption**, and what evidence would prove it wrong.

THE TRAP IN THIS PHASE

The tempting move is to decide what you want to build first, then go hunting through the case study for a justification. It is obvious from the outside every time. **Read first, choose second.** If your rejected problems look like strawmen, the reader will assume you never seriously considered them.

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Phase 2: Build the MVP

Build the smallest thing that actually solves the problem you chose, end to end, for a real user at Dhaga & Co.

Requirement	What it means
A visible frontend	A screen that somebody at Dhaga & Co. can open and use without you standing next to them narrating it. A notebook is not a submission.
Deployed to a URL	Reachable by anyone with the link, running when the room opens it. Hugging Face Spaces with Gradio or Streamlit is the shortest path. Any hosted app platform is fine.
Built in VS Code	Everything lives in the repository and runs locally. Nothing depends on a notebook environment that only exists on one person's machine.
Patterns, chosen on purpose	At least two of prompt chaining, parallelization, routing and evaluator-optimizer, used because the problem needed them. Be ready to say what breaks without each one.
The code and model line	Every step is either deterministic code or a model call. Know which, for every step, and know why. A model earns its place on judgment, language and messy mapping, not on arithmetic, comparison or lookup.
Real-shaped input	It has to hold up on data shaped like what this client actually has: Hinglish, free text, ninety spellings of a colour. Three hand-picked rows is a screenshot, not an MVP.

Vibe-coding is allowed and expected. You are still answerable for every line in the repository, including the lines you did not type.

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Ground rules

Rule	What it means in practice
Two models minimum	Use at least two different models and justify the split on cost, latency and quality. A cheap model doing bulk work with a stronger one on the judgment call is one defensible shape, not the only one.
Cost line	Report what one run costs, and what that becomes at Dhaga & Co.'s actual volume for your use case. Show the arithmetic. The CTO will ask.
Stated temperatures	Extraction, classification and evaluation should not run at the same temperature as anything customer-facing. Say what you chose and why.
Structured output	Every model boundary returns a validated schema, not free text parsed downstream. Handle the case where validation fails.

Rule	What it means in practice
Fails visibly	When your system cannot answer, it says so on screen. Silent wrong answers are worse than visible gaps, and a client will forgive the second.
Cold start in five minutes	A stranger with your README gets it running locally in five minutes. If they cannot, the repository is not finished.

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Phase 3: Present to the client

You are presenting to the client, not to a grader. Assume the room holds Ritu, the function head who owns the problem you picked, and Dev. They are busy, they know their business better than you do, and they have heard confident pitches before.

Segment	Time	What it has to do
The problem	3 min	Name it in their language and show the evidence. They should recognise it as their own problem within the first minute.
Why it matters	3 min	What it costs them today, what improves, and the metric you would hold yourself to.
Live demo	6 min	The deployed MVP, on the real URL, driven live. Include one failure case you show on purpose.
What we would build next	2 min	The version you did not build, what it would need, and what you would want from them to get there.
Questions	6 min	From the room. Expect pushback on cost, on who operates it, and on what happens when it is wrong.

- Every member of the group presents at least one segment.
- Show a failure case. A demo with no failure case is a demo nobody has tested.
- Do not open with your architecture. The client does not care how it works until they care that it works.

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Deliverables

One set per group.

Deliverable	What it has to contain
Discovery note	One page, the seven items listed in Phase 1. Dated before your first commit.
Deployed MVP	A live URL that works when somebody else opens it on a device you have never seen.
Repository	README at the root covering what it does, how to run it, what it expects and what it does when something goes wrong.
Build note	Two pages maximum: the code versus model table, why each pattern is there, the cost line, and the thing that broke that you did not expect.
The presentation	Delivered live, twenty minutes including questions, every member speaking.

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How this is judged

Criterion	What we look for	Weight
Problem selection	A defensible ranking, not just a pick. The problem at the top is owned by a real person and supported by evidence from the case study rather than by enthusiasm.	25
The working MVP	Deployed, usable by someone who did not build it, and holding up on realistic input rather than a curated demo path.	25
Technical judgment	The code and model split is deliberate. Patterns are there because the problem needed them. Cost and failure modes are understood.	20
Client communication	The pitch lands with the room. The problem is recognised, the demo is driven confidently, pushback is handled without defensiveness.	20
What comes next	An honest read of what you did not build, what it would take, and what you would need from the client.	10

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Ways this goes wrong

The failure	Why it costs you
Choosing the build before reading the client	The justification always reads as retrofitted, and the room can tell within a minute.
A problem nobody at the company owns	If you cannot name who loses something today, you have found an idea rather than a problem.
A demo that only works on the happy path	The first question will be what happens when it is wrong. Have an answer you can show.
One builder and four spectators	Every member is asked about the whole system. It is visible immediately.
A frontend that needs narration	If it only makes sense while you explain it, it will not survive contact with the listing team.
Deploying on the morning of the pitch	Deployment is where projects die. Deploy something trivial in the first two days, then keep it alive.
Opening with the architecture	You spend your best three minutes on the part the client cares least about.

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Questions worth sitting with before you start

- 1 Which person in that room loses something if your problem stays unsolved? Are you solving their problem, or the one you found most interesting?
- 2 Dhaga & Co. has sixteen engineers and no ML engineer. Who runs what you built, on the Monday after you leave?
- 3 Your MVP is wrong once in twenty times. Is that acceptable? Whose decision is that, and what changes at once in five?
- 4 If Ritu says "we tried that last year and it did not work", what is your next sentence?
- 5 What is the cheapest thing you could have built that would have tested the same assumption? Why did you not build that instead?
- 6 Which numbers in your pitch came from the case study, and which did you estimate? The room will know the difference, so say which is which before they ask.

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Scope and submission

What we are not asking for

Not production code, not a scalable architecture, not a system integrated with any real Dhaga & Co. platform, and not a polished design. Build the smallest honest thing that proves the idea works and that a real user can operate. An MVP that does one thing convincingly beats four half-built features every time.

Submitting

- One repository per group, with the README at the root and the discovery note and build note inside it.
- The live URL, shared before the presentation day so it can be opened cold.
- All five members present, each covering at least one segment.
- Bring the failure you have not solved yet. The mentor check-in is more useful while the broken parts are still broken.

WHY THIS PROJECT EXISTS

In the field, nobody hands an FDE a specification. They hand you a business, a set of complaints that do not agree with each other, and a limited amount of time before somebody senior wants to see something working. The skill being built here is choosing what to solve, building enough of it to be believed, and making a busy room care.